

LESSON 12.6

PROBLEM-SOLVING WITH VOLUME OF CYLINDERS



LESSON INTRODUCTION

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

LEARNING TARGETS

- I can solve real-world problems involving the volume of right circular cylinders.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

ESSENTIAL QUESTIONS

- What is the relationship between the radius and the height when two right circular cylinders have the same volume?
- How can you use volume to identify missing dimensions in real-world situations?

LESSON NARRATIVE

The lesson continues finding the volume of cylinders by first wondering and noticing about two right cylindrical cans that have the same volume. In 12.6.2 Exploration, students extend their wonder and notice to determine if the volume of a right circular cylinder made from the same piece of paper would vary. The rest of the lesson supports student learning on how to determine the height of a right cylindrical cylinder when given the volume.

COMPONENT SUMMARY

12.6.1 WARM-UP (~5 MIN.)

Compare and contrast two cylindrical cans

12.6.3 GUIDED INSTRUCTION (10 MIN.)

Comparing two cylinders with the same volume but different dimensions to find missing height

12.6.5 YOUR TURN! (10 MIN.)

Students apply what they know about volume in real-world situations

12.6.2 EXPLORATION (15 MIN.)

Exploring the relationship between radius, height, and volume with popcorn containers

12.6.4 TRY IT (5 MIN.)

Practice finding height given volume and radius in real-world context

12.6.6 WRAP-UP (~5 MIN.)

Reflection on lesson learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Calculators
- (Optional) Construction paper
- (Optional) Popcorn

ADDITIONAL PREPARATION

- (Optional) 12.6.2 Exploration includes 3-Act Task Video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may only look for numbers to multiply rather than interpreting the context of a problem. For example, if given volume and radius to find height, students may immediately calculate and multiply those two values rather than making a plan to find the height.
- Students may think that if the volumes are the same that the radius and the height are the same or interchangeable.

THINGS TO CONSIDER

- Consider demonstrating the two views of the lateral area in 12.6.2 Exploration before releasing students to complete the exploration, to ensure that students can interpret the pictures and the questions.
- Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
- If time is an issue, then consider assigning 12.6.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Contemplate, then Calculate

12.6.2 Exploration begins by having students make conjectures. The first two questions serve an important role in student understanding of volume. These conjectures along with 12.6.1 Warm-Up lay the foundation for making meaningful connections between the radius, height, and volume of a cylinder. Encourage students to contemplate and think critically before attempting to calculate an answer.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students must interpret and decode real-world situations involving cylinders throughout Lesson 6. Students first notice and wonder about two cans whose volume is the same. Students then consider how the same lateral area could produce two different volumes. Students then engage in questions within a real-world context to find volume or the height of a cylinder.

DIFFERENTIATE INSTRUCTION

Consider providing visual and kinesthetic entry points in 12.6.2 Exploration activity by providing construction paper for students to create actual popcorn containers.

- Question 1 (Act 1): Students manipulate their pieces of paper to make conjectures about which container holds more popcorn.
- Questions 2-3 (Act 2): Students use mathematics to justify their reasoning by applying dimensions of the container to determine which container has the greater volume. Question 4 (Act 3): Students synthesize their findings. This can also be where students can pour actual popcorn into their containers to verify their findings as well as enjoy the treat.

Alternatively, provide a visual entry point by having students watch Dan Meyer's Three Act Task Popcorn Containers video. An auditory entry point is provided as students make conjectures about which container might hold more popcorn. Finally, students watch the conclusion video to determine their accuracy.

12.6.1 WARM-UP: NOTICE OR WONDER

TEACHER MOVES

Student answers to the Warm-Up should not be checked or corrected. If students have misconceptions during the Warm-Up, those misconceptions should be resolved in 12.6.2 Exploration. Consider prompting students to revisit their answers after 12.6.2 Exploration and provide an opportunity to revise their answers.



12.6.1 Warm-Up: Notice or Wonder

1. Two cylindrical cans are shown.



A close-up picture of each can shows that both cans contain 15 ounces of food.



- a. Explain how it could be possible for two differently shaped cans to have the same volume.

Answers will vary. You can have the same volume if $(\text{radius})^2$ times the height for each can are equal to each other. You can have the same volume if the radius of the first one is smaller than the radius of the second one and the height of the first one is taller than the height of the second one.

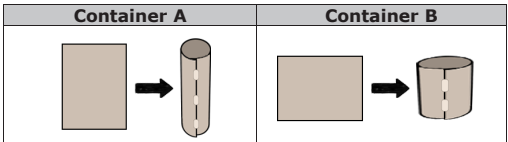
- b. Describe how the dimensions of the two cans compare to each other.

Dimension	How do they compare?
radius	<i>The radius of the asparagus can is smaller than the radius of the chick pea can.</i>
height	<i>The height of the asparagus can is taller than the height of the chick pea can.</i>



12.6.2 Exploration: Popcorn Containers

1. Dan is using paper to create a popcorn container. He can make two differently shaped cylindrical containers by rolling two identical pieces of paper. Container A is taller and narrower, while Container B is shorter and wider.



- a. Which container do you think will hold more popcorn?
Answers will vary. The taller container will hold more popcorn. The shorter wider container will hold more popcorn.
- b. Without having popcorn to test, what information would be helpful in determining which container holds more popcorn?
Answers will vary. The radius and the height of both containers would allow us to determine which holds more.

12.6.2 EXPLORATION: POPCORN CONTAINERS

DIFFERENTIATE INSTRUCTION

Based on your students and classroom, consider a variety of options for implementing this activity.

Option 1: Workbook

Students can engage with each other using their Workbooks to follow the prompts in questions 1- 4.

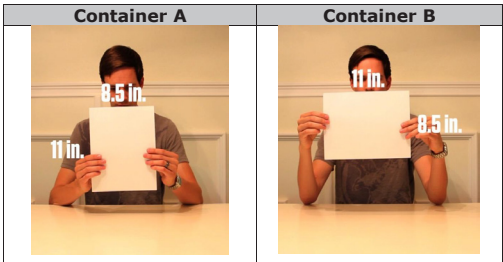
Option 2: Videos

Teacher can play different video portions of Dan Meyer’s Three Act Task that accompanies this activity, playing Task 1 with question 1, Task 2 with questions 2-3, and Task 3 with question 4.

Option 3: Construction Paper Containers

Students can use actual pieces of construction paper and tape to construct and decorate their own popcorn containers. Students follow the prompts in this activity before finishing question 4 by validating their conjectures using real popcorn to pour and enjoy.

The dimensions of each piece of paper prior to rolling it to form a cylinder are shown.



2. The radius of each resulting cylinder is given in the table. Complete the statements.

Container A	Container B
- The height of the paper is the ☐ height ☐ circumference of the cylinder. - The width of the paper is the ☐ height ☐ circumference of the cylinder. - The radius of the cylinder is 1.354 inches (in.).	- The height of the paper is the ☐ height ☐ circumference of the cylinder. - The width of the paper is the ☐ height ☐ circumference of the cylinder. - The radius of the cylinder is 1.752 in.

TEACHER MOVES

Students may need help linking Container A’s shape with Cylinder A and Container B’s shape with Cylinder B. Students may also become confused with the height of the cylinder and the height of the paper. If students prefer to use the length of the paper, then have them change the language in question 2.

DIFFERENTIATE INSTRUCTION

Notice the intentional use of sentence frames in question 2. While circulating the room, use talk moves to prompt students to revoice or reword their sentence frames in their own words to target any misconceptions to later highlight during 12.6.3 Guided Instruction.

12.6.2 EXPLORATION: POPCORN CONTAINERS (CONTINUED)

TEACHER MOVES

Consider allowing students to revisit their responses from 12.6.1 Warm-Up. Guide students to understanding that because the area of the circle is two-dimensional, the dimensions are radius². Then, help them discover that the units of the final answers are three-dimensional because the area is multiplied by the height, a third dimension.

3. Calculate the volume of each container. Use 3.14 for π and round to the nearest thousandth.

Volume of Container A	Volume of Container B
$V = (3.14) (1.354)^2 (11)$ $= 63.323 \text{ in}^3$	$V = (3.14) (1.752)^2 (8.5)$ $= 81.925 \text{ in}^3$

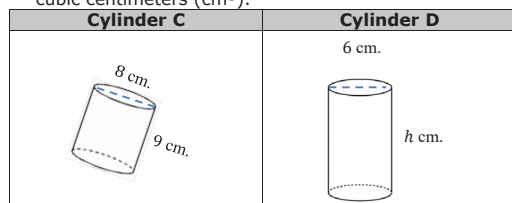
4. Complete the statement.

Container ☐ A ☒ B can hold more popcorn.



12.6.3 Guided Instruction: Using the Volume Formula to Problem-Solve

1. Cylinder C and Cylinder D have the same volume in cubic centimeters (cm^3).



- Find the volume of Cylinder C. Leave it in terms of π .

$$V = \pi \left(\frac{8}{2}\right)^2 (9) = 144\pi$$
- Write an expression for the volume of Cylinder D.

$$V = \pi \left(\frac{6}{2}\right)^2 (h)$$
- Write an equation to show the volumes are equal to each other.

$$\pi \left(\frac{8}{2}\right)^2 (9) = \pi \left(\frac{6}{2}\right)^2 (h)$$
- Solve the equation.

$$\begin{aligned} \pi \left(\frac{8}{2}\right)^2 (9) &= \pi \left(\frac{6}{2}\right)^2 (h) \\ \pi(4)^2(9) &= \pi(3)^2(h) \\ \frac{\pi(4)^2(9)}{\pi(3)^2} &= h \\ \frac{144\pi}{9\pi} &= h \\ h &= 16 \text{ cm} \end{aligned}$$
- What does the solution to the equation represent?
The height of Cylinder D

12.6.3 GUIDED INSTRUCTION: USING THE VOLUME FORMULA TO PROBLEM-SOLVE

TEACHER MOVES

Make connections to the similarities and differences between these two cylinders and the cans from 12.6.1 Warm-Up. Students may better orient to Cylinders C and D through meaningful context like the cans in 12.6.1 Warm-Up.

TEACHER MOVES

Help students articulate that, since their volumes are the same, their equations can be set equal to each other.

QUESTIONING

For students who finish early, challenge them to further explore the mathematical relationship between two cylinders that have the same volume but different dimensions.

- How does r^2 impact the volume?
- What is the relationship between r^2 and h when the volumes are the same?

12.6.4 TRY IT: FINDING THE HEIGHT

DIFFERENTIATE INSTRUCTION

Consider prompting students to sketch the aquarium and its dimensions. They can reference question 2 of 12.6.5 Your Turn! for an example. By sketching the aquarium and its dimensions, students may be less likely to interpret the given volume as the height, thinking they are trying to find volume.



12.6.4 Try It: Finding the Height

1. Julius wants to build an aquarium for his pet goldfish in the shape of a right circular cylinder. He read that his fish needs to live in 1538.6 cubic inches (in.³) of water. If his aquarium has a diameter of 14 in., what is the height of the aquarium? Use 3.14 for π .

$$1538.6 = (3.14) \left(\frac{14}{2} \right)^2 (h)$$

$$1538.6 = 153.86h$$

$$h = 10 \text{ in}$$

12.6.5 YOUR TURN!

TEACHER MOVES

Highlight students who demonstrate flexibility in their work by using a radius of 6 rather than taking half of the diameter. Both are correct in approach, therefore highlighting the flexibility of fluency in problem-solving (MA.K12.MTR.3.1).

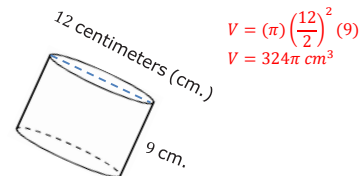
TEACHER MOVES

Students who do not critically engage in decoding the text may assume the phrase “holding 71.5 cubic inches” as the height, therefore multiplying 15.896 times 71.5 rather than what is pictured in the answer key. Prompt students to compare their work to what they did in question 1 of 12.6.4 Try It.



12.6.5 Your Turn!

1. Find the volume of the cylinder shown. Write your answer in terms of π .



$$V = (\pi) \left(\frac{12}{2} \right)^2 (9)$$

$$V = 324\pi \text{ cm}^3$$

2. The ceramic cylinder shown is advertised as holding 71.5 cubic inches. How tall must the ceramic cylinder be? Use 3.14 for π and round your answer to the nearest thousandth.

$$d = 4.5 \text{ in.}$$



$$71.5 = (3.14) \left(\frac{4.5}{2} \right)^2 (h)$$

$$71.5 = 15.896h$$

$$h = 4.498 \text{ in}$$

3. Ruby is building a new above-ground pool for her backyard. The manual says the volume of the pool is 21.613 cubic meters (m³) and has a diameter of 4.6 m. What is the height of the new pool? Use $\frac{22}{7}$ for π and round to the nearest tenth.



$$\begin{aligned} 21.613 &= \left(\frac{22}{7}\right)\left(\frac{4.6}{2}\right)^2 h \\ 21.613 &= 16.6h \\ h &= 1.3 \text{ m} \end{aligned}$$

12.6.5 YOUR TURN! (CONTINUED)

ENRICHMENT

Consider utilizing inquiry to prompt students who need additional support in decoding word problems.

- Mark up the image or sketch your own drawing. What information is provided? What information might you be missing that would be helpful to know?
- What is the question asking you to do?
- Is your answer reasonable, or does it make sense in the context?

 12.6.6 Wrap-Up: Reflection

1. Place an X on the line to indicate your understanding of each learning target. Then, provide a brief explanation or evidence to support your self-assessment.

Answers will vary.

Learning Target	Self-Assessment		
	I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
I can find the volume of a cylinder.			
Evidence			
I can apply the volume formula to solve real-world problems.			
Evidence			

12.6.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

12.6.6 Wrap-Up concludes three consecutive lessons on the volume of cylinders in addition to acting as the final lesson activity in Unit 12. Consider this context when evaluating student responses in 12.6.6 Wrap-Up to inform what intervention and enrichment might look like as students conclude this unit.

Consider prompting students who finish early to try a Desmos extension activity unique to Unit 12.